KAIZEN MATHS

Lesson starter

Indices and Exponent Laws

Index Laws and Operations

Name _____

S1

Simplify: $x^5 \times x^5$

S2

Simplify: $q^6 \div q^3$

S3

Simplify: $(x^3)^3$

S4

Simplify: $a^3 \times a^5$

Show your working

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Exit ticket

Indices and Exponent Laws

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Name _____

E1

Simplify: $n^{10} \div n^5$

E2

Simplify: $(x^4)^3$

E3

Simplify: $b^4 \times b^4$

KAIZEN MATHS

Exit ticket

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Simplify: $n^{10} \div n^5$

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ALG-934A-L2 · E

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Show your working

Teacher answers & working

Indices and Exponent Laws · Index Laws and Operations

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S1

Simplify: $x^5 \times x^5$

Answer:

x^{10}

Working:

Both terms have the same base: x.

$a^m \times a^n = a^{m+n}$

$5 + 5 = 10$

$x^5 \times x^5 = x^{10}$

S2

Simplify: $q^6 \div q^3$

Answer:

q^3

Working:

Both terms have the same base: q.

$a^m \div a^n = a^{m-n}$

$6 - 3 = 3$

$q^6 \div q^3 = q^3$

S3

Simplify: $(x^3)^3$

Answer:

x^9

Working:

Apply the power of a power rule.

$(a^m)^n = a^{m \times n}$

$3 \times 3 = 9$

$(x^3)^3 = x^9$

Teacher answers & working

Indices and Exponent Laws · Index Laws and Operations

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S4

Simplify: $a^3 \times a^5$

Answer:

a^8

Working:

Both terms have the same base: a.

$a^m \times a^n = a^{m+n}$

$3 + 5 = 8$

$a^3 \times a^5 = a^8$

E1

Simplify: $n^{10} \div n^5$

Answer:

n^5

Working:

Both terms have the same base: n.

$a^m \div a^n = a^{m-n}$

$10 - 5 = 5$

$n^{10} \div n^5 = n^5$

E2

Simplify: $(x^4)^3$

Answer:

x^{12}

Working:

Apply the power of a power rule.

$(a^m)^n = a^{m \times n}$

$4 \times 3 = 12$

$(x^4)^3 = x^{12}$

Teacher answers & working

Indices and Exponent Laws · Index Laws and Operations

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E3

Simplify: $b^4 \times b^4$

Answer:

b^8

Working:

Both terms have the same base: b.

$a^m \times a^n = a^{m+n}$

$4 + 4 = 8$

$b^4 \times b^4 = b^8$